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ELECTRONIC DEVICE AND METHOD FOR PROVIDING VIRTUAL IOT ENVIRONMENT

Abstract

An electronic device may comprise: a display, a camera, a memory for storing instructions, and at least one processor, comprising processing circuitry, operatively connected to the display, the camera, and the memory. At least one processor, individually and/or collectively, may be configured to execute the instructions and to cause the electronic device to: activate the camera and display, on the display, an image acquired in real time through the camera; analyze the image to determine at least one space in the image; identify at least one Internet of things (IoT) device which may be placed in the at least one space; and display at least one virtual IoT device corresponding to the identified at least one IoT device in the determined at least one space of the image.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of International Application No. PCT/KR2023/018243 designating the United States, filed on Nov. 14, 2023, in the Korean Intellectual Property Receiving Office and claiming priority to Korean Patent Application Nos. 10-2022-0156704, filed on Nov. 21, 2022, and 10-2022-0170915, filed on Dec. 8, 2022, in the Korean Intellectual Property Office, the disclosures of each of which are incorporated by reference herein in their entireties.

BACKGROUND

Field

[0002] The disclosure relates to an electronic device and, for example, to a method for providing a virtual IoT environment on an electronic device.

Description of Related Art

[0003] The Internet of Things (IoT) may refer to a technology in which various objects are equipped with computing and communication functions, enabling access through the Internet. For example, home appliances such as TVs, refrigerators, washing machines, and vacuum cleaners, various sensors such as temperature sensors and motion sensors, or general devices such as lights and curtains may be utilized as IoT devices. Users may remotely control IoT devices using electronic devices (e.g., smartphones, tablet PCs, and laptop PCs) that support wireless communication functions.

[0004] As IoT technology advances, a large number of IoT devices may be placed in a specific space such as a home, and users may demand additional IoT devices to be utilized in addition to the IoT devices currently in use.

[0005] In the past, there was no technology for recommending IoT devices that can be disposed to enable users to construct an IoT environment in a specific space.

SUMMARY

[0006] An electronic device according to an example embodiment may include: a display, a camera, a memory storing instructions, and at least one processor, comprising processing circuitry, operatively connected to the display, the camera, and the memory.

[0007] According to an example embodiment, at least one processor, individually and/or collectively, may be configured to execute the instructions and to cause the electronic device to: activate the camera and display an image acquired in real time through the camera on the display, analyze the image to determine at least one space within the image, identify at least one IoT (Internet of Things) device capable of being disposed in the at least one space, and display at least one virtual IoT device corresponding to the at least one identified IoT device on the at least one determined space of the image.

[0008] A method for providing a virtual IoT environment of an electronic device according to various example embodiments of the disclosure may include: activating a camera to display an image acquired in real time through the camera, analyzing the image to determine at least one space within the image, identifying at least one IoT (Internet of Things) device capable of being

disposed in the at least one space, and displaying at least one virtual IoT device corresponding to the at least one identified IoT device on the at least one determined space of the image.

[0009] According to various example embodiments of the disclosure, it is possible to provide an electronic device capable of recognizing space and providing a virtual IoT device to a user, and a method for providing a virtual IoT environment of the electronic device.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] The above and other aspects, features and advantages of certain embodiments of the present disclosure will be more apparent from the following detailed description, taken in conjunction with the accompanying drawings, in which:

[0011] FIG. 1 is a block diagram illustrating an example electronic device in a network environment according to various embodiments;

[0012] FIG. 2 is a block diagram illustrating an example configuration of an electronic device according to various embodiments;

[0013] FIG. 3 is a flowchart illustrating an example method for providing a virtual IoT environment of an electronic device according to various embodiments;

[0014] FIG. 4 is a flowchart illustrating an example method for an electronic device to recognize a real object according to various embodiments;

[0015] FIG. 5A and FIG. 5B are diagrams illustrating examples of real IoT devices recognized from a camera image according to various embodiments;

[0016] FIG. 6A and FIG. 6B are diagrams illustrating examples of virtual IoT devices capable of being disposed in a camera image according to various embodiments;

[0017] FIGS. 7A, 7B, 7C and 7D are diagrams illustrating an example camera image and a control item for controlling an IoT device according to various embodiments;

[0018] FIGS. 8A and 8B are diagrams illustrating an example camera image and an affordance item for a virtual IoT device according to various embodiments;

[0019] FIG. 9 is a diagram illustrating a change in a camera image and a control item depending on a change in a camera viewpoint according to various embodiments; and

[0020] FIGS. 10A and 10B are diagrams illustrating an example of recommending a virtual IoT device depending on a real environment according to various embodiments.

DETAILED DESCRIPTION

[0021] FIG. 1 is a block diagram illustrating an example electronic device **101** in a network environment **100** according to various embodiments. Referring to FIG. 1, the electronic device **101** in the network environment **100** may communicate with an electronic device **102** via a first network **198** (e.g., a short-range wireless communication network), or at least one of an electronic device **104** or a server **108** via a second network **199** (e.g., a long-range wireless communication network). According to an embodiment, the electronic device **101** may communicate with the electronic device **104** via the server **108**. According to an embodiment, the electronic device **101** may include a processor **120**, memory **130**, an input module **150**, a sound output module **155**, a display module **160**, an audio module **170**, a sensor module **176**, an interface **177**, a connecting terminal **178**, a haptic module **179**, a camera module **180**, a power management module **188**, a battery **189**, a communication module **190**, a subscriber identification module (SIM) **196**, or an antenna module **197**. In various embodiments, at least one of the components (e.g., the connecting terminal **178**) may be omitted from the electronic device **101**, or one or more other components may be added in the electronic device **101**. In various embodiments, some of the components (e.g., the sensor module **176**, the camera module **180**, or the antenna module **197**) may be implemented as a single component (e.g., the display module **160**).

[0022] The processor **120** may include various processing circuitry and/or multiple processors. For example, as used herein, including the claims, the term “processor” may include various processing circuitry, including at least one processor, wherein one or more of at least one processor, individually and/or collectively in a distributed manner, may be configured to perform various functions described herein. As used herein, when “a processor”, “at least one processor”, and “one or more processors” are described as being configured to perform numerous functions, these terms cover situations, for example and without limitation, in which one processor performs some of recited functions and another processor(s) performs other of recited functions, and also situations in which a single processor may perform all recited functions. Additionally, the at least one processor may include a combination of processors performing various of the recited/disclosed functions, e.g., in a distributed manner. At least one processor may execute program instructions to achieve or perform various functions. The processor **120** may execute, for example, software (e.g., a program **140**) to control at least one other component (e.g., a hardware or software component) of the electronic device **101** coupled with the processor **120**, and may perform various data processing or computation. According to an embodiment, as at least part of the data processing or computation, the processor **120** may store a command or data received from another component (e.g., the sensor module **176** or the communication module **190**) in volatile memory **132**, process the command or the data stored in the volatile memory **132**, and store resulting data in non-volatile memory **134**. According to an embodiment, the processor **120** may include a main processor **121** (e.g., a central processing unit (CPU) or an application processor (AP)), or an auxiliary processor **123** (e.g., a graphics processing unit (GPU), a neural processing unit (NPU), an image signal processor (ISP), a sensor hub processor, or a communication processor (CP)) that is operable independently from, or in conjunction with, the main processor **121**. For example, when the electronic device **101** includes the main processor **121** and the auxiliary processor **123**, the auxiliary processor **123** may be adapted to consume less power than the main processor **121**, or to be specific to a specified function. The auxiliary processor **123** may be implemented as separate from, or as part of the main processor **121**.

[0023] The auxiliary processor **123** may control at least some of functions or states related to at least one component (e.g., the display module **160**, the sensor module **176**, or the communication module **190**) among the components of the electronic device **101**, instead of the main processor **121** while the main processor **121** is in an inactive (e.g., sleep) state, or together with the main processor **121** while the main processor **121** is in an active state (e.g., executing an application). According to an embodiment, the auxiliary processor **123** (e.g., an image signal processor or a communication processor) may be implemented as part of another component (e.g., the camera module **180** or the communication module **190**) functionally related to the auxiliary processor **123**. According to an embodiment, the auxiliary processor **123** (e.g., the neural processing unit) may include a hardware structure specified for artificial intelligence model processing. An artificial intelligence model may be generated by machine learning. Such learning may be performed, e.g., by the electronic device **101** where the artificial intelligence is performed or via a separate server (e.g., the server **108**). Learning algorithms may include, but are not limited to, e.g., supervised learning, unsupervised learning, semi-supervised learning, or reinforcement learning. The artificial intelligence model may include a plurality of artificial neural network layers. The artificial neural network may be a deep neural network (DNN), a convolutional neural network (CNN), a recurrent neural network (RNN), a restricted boltzmann machine (RBM), a deep belief network (DBN), a bidirectional recurrent deep neural network (BRDNN), deep Q-network or a combination of two or more thereof but is not limited thereto. The artificial intelligence model may, additionally or alternatively, include a software structure other than the hardware structure.

[0024] The memory **130** may store various data used by at least one component (e.g., the processor **120** or the sensor module **176**) of the electronic device **101**. The various data may include, for example, software (e.g., the program **140**) and input data or output data for a command related

thereto. The memory **130** may include the volatile memory **132** or the non-volatile memory **134**.

[0025] The program **140** may be stored in the memory **130** as software, and may include, for example, an operating system (OS) **142**, middleware **144**, or an application **146**.

[0026] The input module **150** may receive a command or data to be used by another component (e.g., the processor **120**) of the electronic device **101**, from the outside (e.g., a user) of the electronic device **101**. The input module **150** may include, for example, a microphone, a mouse, a keyboard, a key (e.g., a button), or a digital pen (e.g., a stylus pen).

[0027] The sound output module **155** may output sound signals to the outside of the electronic device **101**. The sound output module **155** may include, for example, a speaker or a receiver. The speaker may be used for general purposes, such as playing multimedia or playing record. The receiver may be used for receiving incoming calls. According to an embodiment, the receiver may be implemented as separate from, or as part of the speaker.

[0028] The display module **160** may visually provide information to the outside (e.g., a user) of the electronic device **101**. The display module **160** may include, for example, a display, a hologram device, or a projector and control circuitry to control a corresponding one of the display, hologram device, and projector. According to an embodiment, the display module **160** may include a touch sensor adapted to detect a touch, or a pressure sensor adapted to measure the intensity of force incurred by the touch.

[0029] The audio module **170** may convert a sound into an electrical signal and vice versa. According to an embodiment, the audio module **170** may obtain the sound via the input module **150**, or output the sound via the sound output module **155** or a headphone of an external electronic device (e.g., an electronic device **102**) directly (e.g., wiredly) or wirelessly coupled with the electronic device **101**.

[0030] The sensor module **176** may detect an operational state (e.g., power or temperature) of the electronic device **101** or an environmental state (e.g., a state of a user) external to the electronic device **101**, and then generate an electrical signal or data value corresponding to the detected state. According to an embodiment, the sensor module **176** may include, for example, a gesture sensor, a gyro sensor, an atmospheric pressure sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a proximity sensor, a color sensor, an infrared (IR) sensor, a biometric sensor, a temperature sensor, a humidity sensor, or an illuminance sensor.

[0031] The interface **177** may support one or more specified protocols to be used for the electronic device **101** to be coupled with the external electronic device (e.g., the electronic device **102**) directly (e.g., wiredly) or wirelessly. According to an embodiment, the interface **177** may include, for example, a high definition multimedia interface (HDMI), a universal serial bus (USB) interface, a secure digital (SD) card interface, or an audio interface.

[0032] A connecting terminal **178** may include a connector via which the electronic device **101** may be physically connected with the external electronic device (e.g., the electronic device **102**). According to an embodiment, the connecting terminal **178** may include, for example, a HDMI connector, a USB connector, a SD card connector, or an audio connector (e.g., a headphone connector).

[0033] The haptic module **179** may convert an electrical signal into a mechanical stimulus (e.g., a vibration or a movement) or electrical stimulus which may be recognized by a user via his tactile sensation or kinesthetic sensation. According to an embodiment, the haptic module **179** may include, for example, a motor, a piezoelectric element, or an electric stimulator.

[0034] The camera module **180** may capture a still image or moving images. According to an embodiment, the camera module **180** may include one or more lenses, image sensors, image signal processors, or flashes.

[0035] The power management module **188** may manage power supplied to the electronic device **101**. According to an embodiment, the power management module **188** may be implemented as at least part of, for example, a power management integrated circuit (PMIC).

[0036] The battery **189** may supply power to at least one component of the electronic device **101**. According to an embodiment, the battery **189** may include, for example, a primary cell which is not rechargeable, a secondary cell which is rechargeable, or a fuel cell.

[0037] The communication module **190** may support establishing a direct (e.g., wired) communication channel or a wireless communication channel between the electronic device **101** and the external electronic device (e.g., the electronic device **102**, the electronic device **104**, or the server **108**) and performing communication via the established communication channel. The communication module **190** may include one or more communication processors that are operable independently from the processor **120** (e.g., the application processor (AP)) and supports a direct (e.g., wired) communication or a wireless communication. According to an embodiment, the communication module **190** may include a wireless communication module **192** (e.g., a cellular communication module, a short-range wireless communication module, or a global navigation satellite system (GNSS) communication module) or a wired communication module **194** (e.g., a local area network (LAN) communication module or a power line communication (PLC) module). A corresponding one of these communication modules may communicate with the external electronic device via the first network **198** (e.g., a short-range communication network, such as Bluetooth™ wireless-fidelity (Wi-Fi) direct, or infrared data association (IrDA)) or the second network **199** (e.g., a long-range communication network, such as a legacy cellular network, a 5G network, a next-generation communication network, the Internet, or a computer network (e.g., LAN or wide area network (WAN))). These various types of communication modules may be implemented as a single component (e.g., a single chip), or may be implemented as multi components (e.g., multi chips) separate from each other. The wireless communication module **192** may identify and authenticate the electronic device **101** in a communication network, such as the first network **198** or the second network **199**, using subscriber information (e.g., international mobile subscriber identity (IMSI)) stored in the subscriber identification module **196**.

[0038] The wireless communication module **192** may support a 5G network, after a 4G network, and next-generation communication technology, e.g., new radio (NR) access technology. The NR access technology may support enhanced mobile broadband (eMBB), massive machine type communications (mMTC), or ultra-reliable and low-latency communications (URLLC). The wireless communication module **192** may support a high-frequency band (e.g., the mmWave band) to achieve, e.g., a high data transmission rate. The wireless communication module **192** may support various technologies for securing performance on a high-frequency band, such as, e.g., beamforming, massive multiple-input and multiple-output (massive MIMO), full dimensional MIMO (FD-MIMO), array antenna, analog beam-forming, or large scale antenna. The wireless communication module **192** may support various requirements specified in the electronic device **101**, an external electronic device (e.g., the electronic device **104**), or a network system (e.g., the second network **199**). According to an embodiment, the wireless communication module **192** may support a peak data rate (e.g., 20 Gbps or more) for implementing eMBB, loss coverage (e.g., 164 dB or less) for implementing mMTC, or U-plane latency (e.g., 0.5 ms or less for each of downlink (DL) and uplink (UL), or a round trip of 1 ms or less) for implementing URLLC.

[0039] The antenna module **197** may transmit or receive a signal or power to or from the outside (e.g., the external electronic device) of the electronic device **101**. According to an embodiment, the antenna module **197** may include an antenna including a radiating element including a conductive material or a conductive pattern formed in or on a substrate (e.g., a printed circuit board (PCB)). According to an embodiment, the antenna module **197** may include a plurality of antennas (e.g., array antennas). In such a case, at least one antenna appropriate for a communication scheme used in the communication network, such as the first network **198** or the second network **199**, may be selected, for example, by the communication module **190** (e.g., the wireless communication module **192**) from the plurality of antennas. The signal or the power may then be transmitted or received between the communication module **190** and the external electronic device via the

selected at least one antenna. According to an embodiment, another component (e.g., a radio frequency integrated circuit (RFIC)) other than the radiating element may be additionally formed as part of the antenna module **197**.

[0040] According to various embodiments, the antenna module **197** may form a mmWave antenna module. According to an embodiment, the mmWave antenna module may include a printed circuit board, a RFIC disposed on a first surface (e.g., the bottom surface) of the printed circuit board, or adjacent to the first surface and capable of supporting a designated high-frequency band (e.g., the mmWave band), and a plurality of antennas (e.g., array antennas) disposed on a second surface (e.g., the top or a side surface) of the printed circuit board, or adjacent to the second surface and capable of transmitting or receiving signals of the designated high-frequency band.

[0041] At least some of the above-described components may be coupled mutually and communicate signals (e.g., commands or data) therebetween via an inter-peripheral communication scheme (e.g., a bus, general purpose input and output (GPIO), serial peripheral interface (SPI), or mobile industry processor interface (MIPI)).

[0042] According to an embodiment, commands or data may be transmitted or received between the electronic device **101** and the external electronic device **104** via the server **108** coupled with the second network **199**. Each of the electronic devices **102** or **104** may be a device of a same type as, or a different type, from the electronic device **101**. According to an embodiment, all or some of operations to be executed at the electronic device **101** may be executed at one or more of the external electronic devices **102**, **104**, or **108**. For example, if the electronic device **101** should perform a function or a service automatically, or in response to a request from a user or another device, the electronic device **101**, instead of, or in addition to, executing the function or the service, may request the one or more external electronic devices to perform at least part of the function or the service. The one or more external electronic devices receiving the request may perform the at least part of the function or the service requested, or an additional function or an additional service related to the request, and transfer an outcome of the performing to the electronic device **101**. The electronic device **101** may provide the outcome, with or without further processing of the outcome, as at least part of a reply to the request. To that end, a cloud computing, distributed computing, mobile edge computing (MEC), or client-server computing technology may be used, for example. The electronic device **101** may provide ultra low-latency services using, e.g., distributed computing or mobile edge computing. In an embodiment, the external electronic device **104** may include an internet-of-things (IoT) device. The server **108** may be an intelligent server using machine learning and/or a neural network. According to an embodiment, the external electronic device **104** or the server **108** may be included in the second network **199**. The electronic device **101** may be applied to intelligent services (e.g., smart home, smart city, smart car, or healthcare) based on 5G communication technology or IoT-related technology.

[0043] FIG. **2** is a block diagram illustrating an example configuration of an electronic device according to various embodiments.

[0044] Referring to FIG. **2**, an electronic device **200** may include a display **210**, a camera **220**, a communication module (e.g., including communication circuitry) **230**, a processor (e.g., including processing circuitry) **250**, and a memory **260**, and may implement various embodiments of the disclosure even if some of the illustrated configurations are omitted or replaced. The electronic device **200** may further include the configurations and/or functions of the electronic device **101** in FIG. **1**.

[0045] According to various embodiments, the electronic device **200** may execute a preview mode of the camera **220** to capture a surrounding environment (e.g., a house), and if there is an IoT (Internet of Things) device connectable in the surrounding environment, operate the identified IoT device using the electronic device **200**, and if there is no connectable IoT device, display a virtual device corresponding to an IoT device, thereby displaying a virtual operation state. The IoT device (or real IoT device) may refer, for example, to an electronic device such as a TV, an IP camera, a

light, or a robot cleaner capable of providing IoT functions.

[0046] According to an embodiment, if there is a connectable IoT device, the electronic device **200** may automatically recognize the device when capturing the device through the camera **220**, establish a connection with the electronic device **200**, and turn on the connected IoT device. In addition, if a user needs to purchase a specific item (or additional components) for the operation of the IoT device, the electronic device may pre-show the operation state in which the item is provided in the camera image (or preview image) through augmented reality (AR).

[0047] According to various embodiments, the display **210** may display an image provided from the processor **250**. The display **210** may be implemented, for example, and without limitation, as one of a liquid crystal display (LCD), a light-emitting diode (LED) display, an organic light-emitting diode (OLED) display, a micro-electromechanical systems (MEMS) display, or an electronic paper display, but is not limited thereto. According to an embodiment, the display **210** may be implemented as a flexible display **210** in which at least a partial area is flexible, and a display area, which is exposed to the outside and displays image information, may vary. For example, the display **210** may be configured as a foldable display or a rollable display. According to an embodiment, the display **210** may include a touch screen that detects a user's touch input. For example, the display **210** may include a capacitive touch screen, but is not limited thereto, and various types of touch screens, such as pressure-sensitive and infrared-type touch screens, may be utilized. The display **210** may include at least some of the configurations and/or functions of the display module **160** in FIG. **1**.

[0048] According to various embodiments, the camera **220** may capture an external environment and provide image information to the processor **250**. The electronic device **200** may include at least one camera on the front side where the display **210** is placed and/or on the back side in the opposite direction, respectively. According to an embodiment, the camera **220** may include a lens assembly including at least one lens for collecting light emitted from an external environment (or a subject), an image sensor (e.g., a charged coupled device (CCD) sensor or a complementary metal oxide semiconductor (CMOS) sensor) for converting light collected through the lens assembly into an electrical signal to generate image data, and an image signal processor for performing various processing on image data acquired from the image sensor. At least some of the above-described configurations of the camera **220** may be omitted or replaced with other configurations. The camera **220** may provide a captured image of an external environment to the processor **250** in real time through an interface (e.g., a mobile industry processor interface). The camera **220** may include at least some of the configurations and/or functions of the camera module **180** in FIG. **1**.

[0049] According to various embodiments, the communication module **230** may include various hardware (e.g., communication circuitry) and/or software elements for supporting wireless communication with an external device. For example, the communication module **230** may communicate with an external device (e.g., an IoT device) located within a predetermined distance through a short-range wireless communication method such as wireless LAN (Wi-Fi) communication, Bluetooth, or ultra-wideband (UWB), and may communicate with an external base station through cellular communication such as 4G LTE (long-term evolution) and 5G NR (new radio). The communication module **230** may support other standard or non-standard wireless communications in addition to the wireless communication methods described above. The communication module **230** may include at least some of the configurations and/or functions of the communication module **190** in FIG. **1**.

[0050] According to various embodiments, the memory **260** may include a known volatile memory and non-volatile memory. The memory **260** may store various instructions that may be executed by the processor **250**. Such instructions may include control commands such as arithmetic and logical operations, data transfer, and input/output, which may be recognized by the processor **250**. The memory **260** may include at least some of the configurations and/or functions of the memory **130** in FIG. **1**, and may store at least some of the programs **140** in FIG. **1**.

[0051] According to various embodiments, the processor **250** may include various processing circuitry capable of performing operations or data processing related to control and/or communication of the respective components of the electronic device **200**, and may include at least some of the configurations of the processor **120** in FIG. **1**. The processor **250** may be operatively, electrically, and/or functionally connected to internal components of the electronic device **200**, such as the display **210**, the camera **220**, the communication module **230**, and the memory **260**. The processor **250** may include various processing circuitry and/or multiple processors. For example, as used herein, including the claims, the term “processor” may include various processing circuitry, including at least one processor, wherein one or more of at least one processor, individually and/or collectively in a distributed manner, may be configured to perform various functions described herein. As used herein, when “a processor”, “at least one processor”, and “one or more processors” are described as being configured to perform numerous functions, these terms cover situations, for example and without limitation, in which one processor performs some of recited functions and another processor(s) performs other of recited functions, and also situations in which a single processor may perform all recited functions. Additionally, the at least one processor may include a combination of processors performing various of the recited/disclosed functions, e.g., in a distributed manner. At least one processor may execute program instructions to achieve or perform various functions.

[0052] According to various embodiments, although the computational and data processing functions capable of being implemented by the processor **250** in the electronic device **200** are not limited, the disclosure will describe various embodiments in which the electronic device **200** analyzes a real-time image acquired from the camera **220** to identify a real IoT device, disposes a virtual IoT device in an empty space on the camera image, controls the operation of the real IoT device and/or the virtual IoT device, and/or provides affordance information associated with the virtual IoT device. The operation of the processor **250** described below may be performed by executing instructions stored in the memory **260**.

[0053] According to various embodiments, the processor **250** may activate the camera **220** and display a camera image acquired in real time from the camera **220** on the display **210**. For example, the camera image may be a preview image, and if the point of view of the camera changes, the surrounding environment displayed through the camera image may change.

[0054] According to various embodiments, the processor **250** may analyze the camera image to recognize at least one object located within the camera image and an empty space where an object is not located. For example, the object may be a real IoT device (e.g., a TV, a light, a robot cleaner, an electric curtain, etc.) capable of providing IoT functions, or a general object (e.g., a general curtain, a sofa, a chair, etc.) that is unable to provide IoT functions. A user may take a picture of a specific space, such as a living room, a room, or an office, using the rear camera of the electronic device **200**. For example, the electronic device **200** may recognize an object located in the camera image and/or an empty space where an object is not located by scanning the space of the camera image.

[0055] According to various embodiments, the processor **250** may recognize a real IoT device from among the objects located in the camera image. According to an embodiment, the processor **250** may recognize a real IoT device by transmitting and receiving a wireless signal using the communication module **230**. For example, the communication module **230** may transmit an advertising packet containing information such as a device name, address, and universally unique identifier (UUID) during a Bluetooth scan process, and an IoT device located at a close distance, upon receiving the advertising packet, may perform a predetermined operation, such as notifying a host device connected to the IoT device or making a request for connection. In addition, the processor **250** may recognize an IoT device located nearby using various short-range wireless communications such as UWB and Wi-Fi through the communication module **230**, and analyze the relative direction and/or distance with respect to the electronic device **200**, thereby determining the

location of the IoT device within the camera image. In addition, the processor **250** may filter out a device transmitting signals of a predetermined intensity or higher using various short-range wireless communications, and register the device transmitting signals of a predetermined intensity or higher. The more types of short-range wireless communications are used, the more accurately the proximity of a device may be recognized through signal intensity and the more accurately the location of the device may be recognized through phase difference. Accordingly, it may be utilized to dispose the virtual device at a more precise position during 3D rendering.

[0056] According to an embodiment, the processor **250** may recognize an object in the camera image through image analysis of the acquired camera image. For example, the processor **250** may include an AI camera (e.g., Bixby vision) capable of detecting various objects included in the camera image acquired in real time. The processor **250** may recognize objects (e.g., a curtain, a light, a stand, etc.) not supporting wireless signal transmission and reception functions, as well as devices capable of being recognized through wireless signal transmission and reception, using the AI camera.

[0057] According to various embodiments, the processor **250** may display a real IoT device recognized within the camera image displayed on the display **210** so as to be recognizable. For example, the processor **250** may display an indicator, indicating the recognized real IoT device, to overlap or adjacent to the real IoT device, highlight the real IoT device, and/or provide a distinct visual effect to an area including the real IoT device.

[0058] According to various embodiments, the processor **250** may display one or more control items respectively corresponding to the recognized real IoT devices on the display **210**. Here, the control item may include at least one button capable of controlling the operation of the real IoT device. When a plurality of real IoT devices are recognized from the camera image, the processor **250** may display control items corresponding to the IoT devices, respectively. According to an embodiment, the processor **250** may display, on the display **210**, the control item in an area separate from the camera image. According to an embodiment, the processor **250** may display the control item in a pop-up form overlaid on the camera image on the display **210**. According to an embodiment, the processor **250** may display a control item in a designated area, based on a user input for an indicator indicating the real IoT device or the virtual IoT device.

[0059] According to various embodiments, the processor **250** may transmit a control command to the real IoT device, based on a user input to the control item. For example, based on a user input to a power button of the control item, the processor **250** may transmit a power-on/off command to a connected TV through the communication module **230**.

[0060] According to various embodiments, the processor **250** may recognize an empty space in the surrounding environment through analysis of the camera image and determine a virtual IoT device capable of being disposed in the empty space. The electronic device **200** may pre-store information related to various types of IoT devices or obtain the information from a cloud server and, if an empty space is identified, select an appropriate IoT device from the pre-stored information and/or a list of IoT devices received from the cloud server.

[0061] According to various embodiments, the processor **250** may display the determined virtual IoT device in the camera image. For example, the virtual IoT device may be a graphic object having substantially the same shape as the corresponding device. The processor **250** may display the virtual IoT device in augmented reality on the surrounding environment of the camera image.

[0062] According to various embodiments, the processor **250** may display a control item corresponding to the virtual IoT device. In the case where real IoT devices exist in the camera image, the processor **250** may display both control items for controlling the respective real IoT devices and control items for controlling respective virtual IoT devices.

[0063] According to various embodiments, the processor **250** may display an execution state of the virtual IoT device in response to a user input. For example, if an input for a start/stop button on a control item of a robot cleaner is detected, the electronic device **200** may generate a real operation

screen, such as movement of the robot cleaner, and display it within the camera image. The processor **250** may display the execution state of the virtual IoT device in augmented reality on the surrounding environment of the camera image. Accordingly, although the virtual IoT device, e.g., the robot cleaner, does not exist in the real environment, the user may predict the operation state when the robot cleaner is disposed in the real environment.

[0064] According to various embodiments, the processor **250** may display an affordance item related to the virtual IoT device. The affordance item may be displayed in a pop-up form adjacent to the virtual IoT device within the camera image, or may be displayed in a separate area from the camera image. The affordance item may include text and/or image information including product information of the virtual IoT device, and a link to a website for purchase of the virtual IoT device. In response to a user input to the affordance item, the processor **250** may access an application or website where the corresponding virtual IoT device is able to be purchased and display a related web page. The user may purchase a real IoT device corresponding to the virtual IoT device from the web page and dispose it in the surrounding environment, thereby utilizing the same. According to an embodiment, the processor **250** may display the affordance item in a designated area, based on a user input to an indicator indicating the real IoT device or the virtual IoT device.

[0065] FIG. **3** is a flowchart illustrating an example method for providing a virtual IoT environment of an electronic device according to various embodiments.

[0066] The method illustrated in FIG. **3** may be performed by an electronic device (e.g., the electronic device **200** in FIG. **2**), and the technical features described above may not be repeated in the description below.

[0067] According to various embodiments, in operation **310**, the electronic device may activate a camera (e.g., the camera **220** in FIG. **2**) and display a camera image (e.g., a preview image) acquired in real time from the camera on a display (e.g., the display **210** in FIG. **2**).

[0068] According to various embodiments, in operation **320**, the electronic device may analyze the camera image and scan the space within the camera image. The user may take a picture of a specific space, such as a living room, a room, or an office, using the rear camera of the electronic device. For example, the electronic device may recognize an object located within the camera image and/or an empty space where an object is not located by scanning the space of the camera image. The electronic device may perform operations **330** to **345**, based on a result of recognizing the object, and may perform operations **350** to **390**, based on a result of recognizing the empty space. Operations **330** to **345** and operations **350** to **390** may be performed in parallel with each other, and at least some of the operations may be performed simultaneously.

[0069] According to various embodiments, in operation **330**, the electronic device may recognize a real IoT device located within the camera image.

[0070] According to an embodiment, the electronic device may recognize a real IoT device through transmission and reception of a wireless signal. For example, the electronic device may transmit an advertising packet containing information such as a device name, address, and universally unique identifier (UUID) during a Bluetooth scan process, and an IoT device located at a close distance, upon receiving the advertising packet, may perform a predetermined operation, such as notifying a host device connected to the IoT device or making a request for connection. In addition, the processor **250** may recognize an IoT device located nearby using various short-range wireless communications such as UWB and Wi-Fi, and analyze the relative direction and/or distance with respect to the electronic device, thereby determining the location of the IoT device within the camera image. In addition, the electronic device may filter out a device transmitting signals of a predetermined intensity or higher using various short-range wireless communications, and register the device transmitting signals of a predetermined intensity or higher. The more types of short-range wireless communications are used, the more accurately the proximity of a device may be recognized through signal intensity and the more accurately the location of the device may be recognized through phase difference. Accordingly, it may be utilized to dispose the virtual device at

a more precise position during 3D rendering.

[0071] According to an embodiment, the electronic device may recognize an object within the camera image through image analysis of the acquired camera image. For example, the electronic device may include an AI camera (e.g., Bixby vision) capable of detecting various objects included in the camera image acquired in real time. The electronic device may recognize objects (e.g., a curtain, a light, a stand, etc.) not supporting wireless signal transmission and reception functions, as well as devices capable of being recognized through wireless signal transmission and reception, using the AI camera.

[0072] According to an embodiment, the electronic device may establish a connection with the recognized real IoT device. For example, if a connection with a specific IoT device is not currently established, the electronic device may establish a connection through short-range wireless communication (e.g., Wi-Fi) and register connection information with a cloud server that provides IoT services.

[0073] According to various embodiments, in operation **335**, the electronic device may display the real IoT device recognized within the camera image so as to be recognizable. For example, the electronic device may display an indicator, indicating the recognized real IoT device, to overlap or adjacent to the real IoT device, highlight the real IoT device, and/or provide a distinct visual effect to an area including the real IoT device.

[0074] According to an embodiment, the electronic device may display one or more control items respectively corresponding to the recognized real IoT devices on the display. The electronic device may display a control item area including at least one control item separately from an area where the camera image is displayed (e.g., at the bottom of the camera image), and as will be described below, the control item area may include at least one control item corresponding to the real IoT device and at least one control item corresponding to the virtual IoT device.

[0075] According to various embodiments, in operation **340**, the electronic device may receive an input (e.g., a user input) to the control item. For example, a control item corresponding to a TV, as a real IoT device, may include a power button, a channel control button, and/or a volume control button, and the user input may be an input to a specific button of the control item.

[0076] According to various embodiments, in operation **345**, the electronic device may transmit an operation control command for the real IoT device in response to the user input to the control item. For example, if the user selects a TV power button on the control item, a TV on/off command may be directly transmitted to the TV through the established short-range wireless communication, and/or transmitted to the TV via a cloud server on the network.

[0077] According to various embodiments, in operation **350**, the electronic device may recognize an empty space in the camera image. For example, the electronic device may recognize an object or the space in the camera image through transmission/reception of a wireless signal and/or image analysis using an AI camera, and may recognize a space having a predetermined size or more in which an object is not recognized as an empty space.

[0078] According to various embodiments, in operation **355**, the electronic device may determine a virtual IoT device capable of being disposed in the identified empty space. The electronic device may pre-store information related to various types of IoT devices or obtain information from a cloud server and, if an empty space is identified, select an appropriate IoT device from the pre-stored information and/or a list of IoT devices received from the cloud server.

[0079] According to an embodiment, the electronic device may recognize a pattern of the recognized spaces, extract their features, and determine virtual IoT devices suitable for the spaces. For example, if the environment being captured through the camera image is messy with many objects and few empty spaces, the electronic device may select a device that is helpful for organizing and has a small volume (e.g., a handheld cleaner). If the environment being captured is in a well-organized state with many objects and few empty spaces, the electronic device may select an additional device that is convenient and helpful for organizing the surrounding environment

better or a device having a small volume (e.g., an air purifier, a water purifier, an air dresser, a smart picture frame, etc.). If the environment being captured is messy with few objects and many empty spaces, the electronic device may select a device that is helpful for organization and has a large volume (e.g., a robot cleaner, a storage closet, etc.). If the environment being captured is in a well-organized state with few objects and many empty spaces, the electronic device may select an additional device that is convenient and helpful for organizing the surrounding environment better or a device having a large volume (e.g., a robot cleaner, a smart speaker, etc.).

[0080] According to various embodiments, in operation **360**, the electronic device may display the determined virtual IoT device in the camera image. The electronic device may display the virtual IoT device in an augmented reality on the surrounding environment of the camera image.

[0081] According to an embodiment, the electronic device may display the virtual IoT device so as to be distinguished from the objects including the real IoT devices in the camera image. The electronic device may display an indicator, indicating the virtual IoT device, adjacent to or overlapping the virtual IoT device, and display a control item corresponding to the virtual IoT device.

[0082] According to various embodiments, in operation **370**, the electronic device may receive an input (e.g., a user input) to the control item corresponding to the virtual IoT device. For example, if the selected virtual IoT device is a robot cleaner, the control item may include a start/stop button of the robot cleaner.

[0083] According to various embodiments, in operation **375**, the electronic device, in response to the user input, may display the operation state of the virtual IoT device. The electronic device may display the operation state of the virtual IoT device in augmented reality on the surrounding environment of the camera image. For example, if an input to a start/stop button on the control item of the robot cleaner is detected, the electronic device may generate a real operation screen, such as movement of the robot cleaner, and display it in the camera image. Accordingly, although the robot cleaner, which is a virtual IoT device, does not exist in the real environment, the user may predict the operation state when the robot cleaner is disposed in reality.

[0084] According to various embodiments, in operation **380**, the electronic device may display an affordance item related to the virtual IoT device. The affordance item may be displayed in a pop-up form adjacent to the virtual IoT device in the camera image, or may be displayed in a separate area from the camera image. The affordance item may include text and/or image information including product information of the virtual IoT device, and a link to a website for purchase of the virtual IoT device.

[0085] According to various embodiments, in operation **385**, the electronic device may receive a user input to the affordance item. According to various embodiments, in operation **390**, the electronic device may access a website where the corresponding virtual IoT device is able to be purchased in response to the user input and display a related web page.

[0086] FIG. **4** is a flowchart illustrating an example method for an electronic device to recognize a real object according to various embodiments.

[0087] The method illustrated in FIG. **4** may be performed by an electronic device (e.g., the electronic device **200** in FIG. **2**), and the description of the technical features described above may not be repeated here.

[0088] According to various embodiments, in operation **410**, the electronic device may activate a camera (e.g., the camera **220** in FIG. **2**) to execute a preview mode, and display a camera image (or a preview image) acquired in real time on a display (e.g., the display **210** in FIG. **2**). The user may capture a specific space, such as a living room, a room, or an office, using a rear camera of the electronic device, and the captured camera image may include a plurality of objects that belong or do not belong to IoT devices.

[0089] According to various embodiments, in operation **420**, the electronic device may recognize a short-range wireless communication signal transmitted from an adjacent IoT device. According to

an embodiment, the electronic device may support short-range wireless communication, such as Wi-Fi, Bluetooth (or Bluetooth low energy (BLE)), and ultra-wideband (UWB). The electronic device may recognize an adjacent IoT device by transmitting and receiving a short-range wireless communication signal. For example, the electronic device may transmit an advertising packet containing information such as a device name, address, and universally unique identifier (UUID) during a Bluetooth scan process, and an IoT device located at a close distance, upon receiving the advertising packet, may perform a predetermined operation, such as notifying a host device connected to the IoT device or making a request for connection. In addition, the electronic device may recognize an IoT device located nearby using various short-range wireless communications such as UWB and Wi-Fi, and analyze the relative direction and/or distance with respect to the electronic device, thereby determining the location of the IoT device within the camera image. In addition, the electronic device may filter out a device transmitting signals of a predetermined intensity or higher using various short-range wireless communications, and register the device transmitting signals of a predetermined intensity or higher. The more types of short-range wireless communications are used, the more accurately the proximity of a device may be recognized through signal intensity and the more accurately the location of the device may be recognized through phase difference. Accordingly, it may be utilized to dispose the virtual device at a more precise position during 3D rendering.

[0090] According to various embodiments, in operation **430**, the electronic device may recognize at least one object included in the camera image using an AI camera. For example, the electronic device may include an AI camera (e.g., Bixby vision) capable of detecting various objects included in the camera image acquired in real time. The electronic device may recognize objects (e.g., a curtain, a light, a stand, etc.) not supporting wireless signal transmission and reception functions, as well as devices capable of being recognized through wireless signal transmission and reception, using the AI camera. According to various embodiments, operations **420** and **430** may be performed sequentially or at least partially simultaneously.

[0091] According to various embodiments, in operation **440**, the electronic device may recognize an object displayed within the camera image. For example, the electronic device may recognize at least one object included in the camera image through wireless signal transmission and reception in operation **420** and/or object recognition using the AI camera in operation **430**. The recognized object may include an IoT device equipped with an IoT function, such as a TV, an air conditioner, or a robot cleaner, an object, such as a light or curtain, that is not currently equipped with an IoT function but to which an IoT function may be added through a hub, and/or an object that is unable to provide an IoT function, such as a flowerpot or a sofa.

[0092] According to various embodiments, in operation **450**, the electronic device may recognize an object belonging to the category of IoT devices from among the objects recognized in the camera image. For example, the electronic device may recognize an object belonging to a list of IoT devices stored in the memory and/or a database of a cloud server, from among the objects recognized in operation **440**, as an IoT device. According to an embodiment, the electronic device may recognize a device to which IoT functions is able to be added from the database, among general objects (e.g., curtains, lights, stands, etc.) that do not support wireless signal transmission and reception, so as to distinguish it from other objects.

[0093] FIG. 5A and FIG. 5B are diagrams illustrating examples of real IoT devices recognized from a camera image according to various embodiments.

[0094] FIGS. 5A and 5B illustrate camera images captured in real time by a camera (e.g., the camera **220** in FIG. 2) of an electronic device (e.g., the electronic device **200** in FIG. 2) and displayed on a display (e.g., the display **210** in FIG. 2).

[0095] Referring to FIGS. 5A and 5B, a TV **521**, which is an IoT device capable of connecting with an electronic device via wireless communication, a light **561**, which is an object capable of providing an IoT function through connection with an IoT hub, and a curtain capable of providing

an IoT function by installing additional components (or items) may be located within the surrounding environment being captured. According to an embodiment, an electronic device (e.g., the electronic device **200** in FIG. 2) may recognize one or more objects located within a camera image **510** through recognition of a short-range wireless communication signal received from an external device and/or image analysis of an AI camera (e.g., Bixby vision), and recognize a real IoT device from among the recognized objects. For example, among the TV **521**, the light **561**, and the curtain, which are objects recognized in FIG. 5A, the electronic device may recognize the TV **521**, which belongs to the IoT device list of the database, as a real IoT device.

[0096] According to an embodiment, the electronic device may display an indicator **520** indicating the identified real IoT device (e.g., the TV **521**) within the camera image **510**. The indicator **520** may be displayed overlapping and/or adjacent to the real IoT device.

[0097] According to an embodiment, the electronic device may display a control item **525** capable of controlling the operation of the identified real IoT device. For example, the control item **525** corresponding to the real IoT device, e.g., the TV **521**, may include a power button, a channel control button, and/or a volume control button. According to an embodiment, as illustrated in FIGS. 7 and 8, the electronic device may display at least one control item in an area distinct from the camera image. According to an embodiment, the electronic device may display the control item **525** in a pop-up form in an area of the camera image **510**.

[0098] According to an embodiment, the electronic device may transmit an operation control command of the real IoT device in response to a user input to the control item **525**. For example, if the user selects a power button of the TV **521** on the control item **525**, the on/off command for the TV **521** may be directly transmitted to the TV **521** through a short-range wireless communication established between the TV **521** and a communication module (e.g., the communication module **230** in FIG. 2), and/or transmitted to the TV **521** through a cloud server on the network. The TV **521** may be turned on according to the control command received from the electronic device. Accordingly, the user may identify that the real TV **521** is turned on in the camera image **530**.

[0099] According to an embodiment, the electronic device may provide an affordance item **540** corresponding to the identified real IoT device. Referring to FIG. 5A, the affordance item **540** may include text and/or image information about additional functions that may be further provided through the real IoT device, and may include a link **545** for access to a website that provides additional functions or purchases of additional products. If the user selects the link **545** in the affordance item **540**, the linked website may be displayed, and the user may obtain additional information about the IoT device or purchase additional products from the website.

[0100] Referring to FIG. 5B, the light **561** located within the environment being captured may be an object that is connected to an IoT hub to provide IoT functions but not yet connected to the electronic device.

[0101] According to an embodiment, the electronic device may display an indicator **560** corresponding to the identified object, e.g., the light **561**, within the camera image **550**. For example, the electronic device may display the indicator **560** adjacent to or at least partially overlapping the object, e.g., the light **561**. The electronic device may display a control item **565** corresponding to the light **561** and, based on a user input to the control item **565**, transmit a control command to an IoT hub connected to the light **561** to turn on the light **561**. According to an embodiment, if a touch input to the indicator **560** is detected, the electronic device may display the control item **565** on an area separated from the camera image **550** or display it in a pop-up form on at least a portion of the camera image **550**.

[0102] According to an embodiment, the electronic device may display an affordance item **580** corresponding to the object. For example, the affordance item **580** corresponding to the light **561**, which is an object that is not yet connected to the electronic device, may include text and/or image information, indicating that it is connectable, and a link **585** enabling access to an application or website capable of performing the connection. If the user selects the link **585** of the affordance item

580, the electronic device may attempt to connect and register with a hub connected to the light. According to an embodiment, if a touch input to the indicator **560** is detected, the electronic device may display the affordance item **580** on an area separated from the camera image **550** or display it in a pop-up form on at least a portion of the camera image **550**.

[0103] According to an embodiment, if the electronic device is connected to the light (or the hub connected to the light), a control item corresponding to the light may be displayed. Thereafter, in response to a user input to the control item, the electronic device may transmit a control command to the light, and accordingly, the light **561** may be turned on, as illustrated in FIG. 5B.

[0104] FIG. 6A and FIG. 6B are diagrams illustrating examples of virtual IoT devices capable of being disposed in a camera image according to various embodiments.

[0105] According to various embodiments, an electronic device (e.g., the electronic device **220** in FIG. 2) may determine at least one space in a camera image, identify an IoT device capable of being disposed in the determined space, and display a virtual IoT device corresponding to the identified IoT device in the determined space within the image. The virtual IoT device may be a graphic object in the form of a real IoT device.

[0106] Referring to FIG. 6A, the electronic device may recognize a space between a TV and a curtain as an empty space in a camera image **610** captured in real time. The electronic device may determine a robot cleaner as an IoT device capable of being disposed in the empty space, and display a virtual IoT device **621**, which is a graphic object in the form of a robot cleaner, in the camera image. The electronic device may display an indicator **620**, indicating the virtual IoT device **621**, adjacent to the virtual IoT device **621**.

[0107] According to an embodiment, the electronic device may display a control item **625** capable of controlling the virtual operation of the virtual IoT device **621**. The control item **625** corresponding to the virtual IoT device **621** may be substantially the same as a control item (e.g., the control item **525** in FIG. 5A) corresponding to the real IoT device.

[0108] According to an embodiment, the electronic device, based on a user input to the control item **625**, may display a virtual operation state **622** corresponding to the operation state of the virtual IoT device **621** during operation within the camera image **630**. For example, if a start button for a robot cleaner is pressed, the robot cleaner may move to another space in the house. The virtual operation state **622** may show the same effect as the real robot cleaner moving as a movement of a graphic object. Accordingly, the user may predict the real operation state when purchasing and installing the corresponding IoT device through the image **630** in which the real environment and the operation state of the virtual IoT device **621** overlap each other.

[0109] According to an embodiment, the electronic device may provide an affordance item **640** corresponding to the virtual IoT device **621**. Referring to FIG. 6A, it may include text and/or image information including related information of a robot cleaner corresponding to the virtual IoT device **621** and a link **645** to an application or website for purchasing the robot cleaner. If the user selects the link of the affordance item **640**, the electronic device may access the application or website for purchasing the robot cleaner and provide a corresponding screen thereof to the user through the display.

[0110] Referring to FIG. 6B, among the objects recognized from the camera image **650**, a curtain **661** may be an object that may provide an IoT function through the installation of an additional product. For example, an electric curtain with an automatic opening/closing device added, unlike a curtain in the real environment, may provide a function to automatically open and close the electric curtain through an input onto the electronic device.

[0111] According to an embodiment, the electronic device may display an indicator **660** corresponding to the recognized object (e.g., the electric curtain) and a control item **665** capable of controlling the electric curtain. The electronic device, based on a user input to the control item, may display a virtual operation state **662** of the electric curtain, which is a virtual IoT device. For example, if the user selects a close button on the control item **665**, the electronic device may

display a virtual operation state **662** providing an effect similar to opening the curtain within the camera image **670**.

[0112] According to an embodiment, the electronic device may provide an affordance item **680** corresponding to the virtual IoT device. Referring to FIG. **6B**, it may include text and/or image information including information of the electric curtain corresponding to the virtual IoT device and a link **685** to an application or website for purchasing the electric curtain. If the user selects the link **685** of the affordance item **680**, the electronic device may access an application or website for purchasing the electric curtain and provide a corresponding screen thereof to the user through the display.

[0113] FIGS. **7A**, **7B**, **7C** and **7D** are diagrams illustrating example camera images and a control item for controlling an IoT device according to various embodiments.

[0114] According to various embodiments, an electronic device **700** (e.g., the electronic device **200** in FIG. **2**) may display a camera image **710** (or a preview image) acquired in real time through a camera (e.g., the camera **220** in FIG. **2**) on a first area (or a camera image display area) of a display **701** (e.g., the display **210** in FIG. **2**).

[0115] The electronic device **700** may display control items **731**, **733**, **735**, and **737** corresponding to respective real IoT devices recognized from the camera image **710** and/or virtual IoT devices disposed in the space of the camera image **710** on a second area **760** (or a control item area) of the display **701**. Referring to FIG. **7**, although the camera image **710** and the control items are displayed on the first area at the top and the second area **760** at the bottom of the display **701**, respectively, the locations thereof are not limited thereto.

[0116] Referring to FIGS. **7A**, **7B**, **7C** and **7D** (which may be referred to collectively as FIG. **7**), the image **710** of the surrounding environment captured using the camera of the electronic device **700** may include various objects, and for example, a TV **721**, a light **723**, and a curtain **727** may be recognized.

[0117] According to an embodiment, the electronic device **700** may recognize one or more objects located in the camera image **710** through the recognition of a short-range wireless communication signal and/or the image analysis of an AI camera (e.g., Bixby vision), and recognize real IoT devices from among the recognized objects. For example, the electronic device **700** may recognize the TV **721** and the light **723** belonging to the IoT device list of the database, among the objects, such as the TV **721**, the light **723**, and the curtain **727**, recognized in FIG. **7**, as real IoT devices.

[0118] According to an embodiment, the electronic device **700** may determine an IoT device to be disposed in an empty space within the camera image **710** and display a virtual IoT device corresponding to the determined IoT device in the camera image **710**. For example, the electronic device **700** may determine a robot cleaner **725** as an IoT device capable of being disposed in the space between the TV **721** and the curtain **727**, and an electric curtain capable of replacing the curtain **727**, and display a virtual IoT device **727** corresponding to the electric curtain.

[0119] According to an embodiment, the electronic device **700** may display control items **731**, **733**, **735**, and **737** respectively corresponding to the real IoT devices and/or the virtual IoT devices in the second area of the display **701**. For example, referring to FIG. **7**, a control item **731** capable of controlling the TV **721** as a real IoT device, a control item **733** capable of controlling the light **723**, a control item **735** capable of controlling the robot cleaner **725** as a virtual IoT device, and a control item **737** capable of controlling the electric curtain **727** may be displayed. Each of the control items **731**, **733**, **735**, and **737** may include image/text information indicating an IoT device, a current state, and/or at least one button capable of controlling the operation of the device.

According to an embodiment, the electronic device **700** may provide an integrated control item **738** capable of controlling a plurality of IoT devices at once. For example, if Good-morning button of the integrated control item **738** is selected, control may be performed such that the electric curtain **727** is opened, the light **723** is turned off, and the robot cleaner **725** is operated.

[0120] According to an embodiment, if a real IoT device is recognized from the camera image or if

a virtual IoT device is determined through image analysis, the electronic device **700** may display a control item corresponding to the real IoT device or the virtual IoT device on one area of the display. According to an embodiment, based on a user input to an indicator indicating the real IoT device or the virtual IoT device, the electronic device **700** may display a control item on one area of the display.

[0121] According to an embodiment, the electronic device **700** may transmit an operation control command of the real IoT device, based on a user input to the control item corresponding to the real IoT device. In FIG. 7A, if the user selects a close button on the control item **737** of the electric curtain **727**, the electronic device **700** may transmit a control command to close the electric curtain **727**. In addition, in FIG. 7B, if the user selects a turn-on button on the control item **733** of the light **723**, the electronic device **700** may transmit a control command to turn on the light **723**.

[0122] According to an embodiment, based on a user input to the control item of the virtual IoT device, the electronic device **700** may display a virtual operation state corresponding to an operation state of the IoT device during operation in the camera image **710**. In FIG. 7C, if the user selects a start button of the control item **735** of the robot cleaner **725**, which is a virtual IoT device, as shown in FIG. 7D, the electronic device **700** may display, within the camera image **710**, a virtual operation state **740** including movement of a graphic object showing the same effect as the real robot cleaner **725** moving.

[0123] FIGS. **8A** and **8B** are diagrams illustrating examples of a camera image and an affordance item for a virtual IoT device according to various embodiments.

[0124] According to various embodiments, the electronic device **800** may provide an affordance item **880** corresponding to a virtual IoT device further displayed on an empty space in a camera image **810**. Referring to FIGS. **8A** and **8B** (which may be referred to collectively as FIG. **8**), the electronic device **800** may generate a virtual object corresponding to an electric curtain **827** as a virtual IoT device and display it within the camera image **810**.

[0125] Referring to FIG. **8**, the electronic device **800** may display the camera image **810** on a first area of a display **801** and display control items **831**, **833**, **835**, and **837** respectively corresponding to real IoT devices and virtual IoT devices on a second area of the display **801**. In FIG. **8A**, if the user presses a close button of the electric curtain as a virtual IoT device, the electronic device **800** may implement a virtual operation state showing the same effect as the electric curtain closing within the camera image **810**, as shown in FIG. **8B**.

[0126] According to an embodiment, the electronic device **800** may provide an affordance item **880** corresponding to the virtual IoT device. For example, the affordance item **880** may include text and/or image information including information of the electric curtain corresponding to the virtual IoT device and a link **885** to an application or website for purchasing the electric curtain. If the user selects the link **885** of the affordance item **880**, the electronic device **800** may access an application or website for purchasing electric curtains and provide a corresponding screen thereof to the user through the display **801**.

[0127] FIG. **9** is a diagram illustrating an example change in a camera image and a control item depending on a change in a camera viewpoint according to various embodiments.

[0128] According to an embodiment, when the viewpoint of a camera (e.g., the camera **220** in FIG. **2**) is changed so that a real IoT device displayed is added to be displayed on the image, an electronic device **900** (e.g., the electronic device **200** in FIG. **2**) may recognize the added real IoT device and display a control item of the further recognized real IoT device. In addition, if the viewpoint of the camera is changed so that the real IoT device displayed in the image disappears, the electronic device **900** may recognize that the real IoT device has disappeared and remove the control item of the corresponding real IoT device.

[0129] According to an embodiment, when the viewpoint of the camera is changed so that the empty space in the camera image disappears, the electronic device **900** may remove the virtual IoT device and control item that are determined to be disposed in the empty space. [0130] (A) of FIG. **9**

illustrates a state in which the camera is capturing the left side of the surrounding environment.

From the viewpoint (A) of FIG. 9, an electric curtain **971** may be recognized as a real IoT device in the camera image. In addition, the electronic device **900** may determine an empty space capable of disposing an IoT device in the camera image and a virtual IoT device capable of being disposed in the empty space. Referring to (A) in FIG. 9, a robot cleaner may be determined to be disposed in the lower space **980**. The electronic device **900** may display a control item **991** of the electric curtain, which is the recognized real IoT device, and a control item **992** of the robot cleaner, which is the virtual IoT device, in a control item area **960** on a display **901**. [0131] (B) of FIG. 9

illustrates a state in which the camera is capturing the center of the surrounding environment. From the viewpoint (B) of FIG. 9, a light **973** and an electric curtain **971** may be recognized as real IoT devices from the camera image. In addition, the electronic device **900** may determine that the robot cleaner is able to be disposed in the lower space **982**. The electronic device **900** may further display a control item **993** of the light **973**, which is the real IoT device further recognized, on the display **901**. [0132] (C) of FIG. 9 illustrates a state in which the camera is capturing the right side of the surrounding environment. From the viewpoint (C) of FIG. 9, the light **973** and the electric curtain **971** may be recognized as real IoT devices from the camera image. In addition, the electronic device **900** may determine that a TV is able to be disposed as a virtual IoT device in the right space **984**. The electronic device **900** may display a virtual object of the TV within the camera image and further display a control item **994** of the TV. In addition, the space **982** for disposing the robot cleaner recognized in (A) and (B) in FIG. 9 may not be visible depending on the change in the viewpoint of the camera. The electronic device **900** may remove the virtual object and control item **992** of the robot cleaner depending on the change in the viewpoint.

[0133] FIGS. 10A and 10B are diagrams illustrating an example of recommending a virtual IoT device depending on a real environment according to various embodiments.

[0134] FIGS. 10A and 10B illustrate different surrounding environments. The electronic device may analyze objects and empty spaces in the camera image, and determine virtual IoT devices capable of being disposed in the empty spaces depending on the properties of the surrounding environment. For example, multiple empty spaces **1023**, **1025**, and **1027** may be recognized within the environment being captured, as shown in FIG. 10A, or an empty space may be recognized only in a narrow area **1071** below within the environment being captured, as shown in FIG. 10B.

[0135] According to an embodiment, if the environment being captured through the camera image **1010** is messy with many objects and few empty spaces, a device (e.g., a handy cleaner) helpful for organization and having a small volume may be selected.

[0136] According to an embodiment, if the environment being captured is in a well-organized state with many objects and few empty spaces, an additional device (e.g., an air purifier, a water purifier, an air dresser, a smart picture frame, etc.) that is convenient and helpful for organizing the surrounding environment better or a device having a small volume may be selected.

[0137] In an embodiment, if the environment being captured is messy with few objects and many empty spaces, a device (e.g., a robot cleaner, a storage closet, etc.) that is helpful for organization and has a large volume may be selected. Alternatively, if the environment being captured is in a well-organized state with few objects and many empty spaces, an additional device that is convenient and helpful for organizing the surrounding environment better or a device having a large volume (e.g., a robot cleaner, a smart speaker, etc.) may be selected.

[0138] According to an embodiment, the electronic device may dispose the virtual IoT device in the empty space of the surrounding environment within the camera image or display it overlapping other objects. The electronic device may preferentially dispose the virtual IoT device in the empty space and, if there is only a small empty space in the surrounding environment, dispose the virtual IoT device to overlap the empty space and surrounding objects, and, if there is no empty space, dispose the IoT device to overlap other objects at an optimal location.

[0139] According to an embodiment, if there is no connectable real IoT device in the surrounding

environment captured in the camera image and if a new IoT device needs to be purchased, the electronic device may recommend and display a virtual IoT device so that the user may experience it in advance. In addition, if there is no connectable real IoT device in the surrounding environment captured in the camera image and if provision of additional components enables an IoT function while being connected with the general device, the electronic device may induce the purchase of additional components for IoT execution.

[0140] According to an embodiment, if there is a connectable real IoT device in the surrounding environment captured in the camera image and if the IoT device has never been connected to the electronic device and is in an off state, the electronic device may recommend to turn on the IoT device to operate and to use the same after connection. In addition, if the IoT device has never been connected to the electronic device and is in an on state, the electronic device may recommend inducing connection so as to operate, connect, and then use the IoT device. In addition, if the IoT device has ever been connected to the electronic device and is currently in an off state, the electronic device may induce to turn on the IoT device to operate and then continuously use the same. In addition, if the IoT device has ever been connected to the electronic device and is currently in an on state, the electronic device may induce to continuously use the same and provide recommendations about other IoT devices that would be good to use together.

[0141] An electronic device according to various example embodiments of the disclosure may include a display, a camera, a memory storing instructions, and at least one processor, comprising processing circuitry, operatively connected to the display, the camera, and the memory.

[0142] According to an example embodiment, at least one processor, individually and/or collectively, may be configured to execute the instructions and to cause the electronic device to: activate the camera and display an image acquired in real time through the camera on the display, analyze the image to determine at least one space within the image, identify at least one Internet of Things (IoT) device capable of being disposed in the at least one space, and display at least one virtual IoT device corresponding to the at least one identified IoT device on the at least one determined space of the image.

[0143] According to an example embodiment, at least one processor, individually and/or collectively, may be configured to cause the electronic device to: identify at least one real object included in the image through analysis of the image and identify an IoT device that is connectable with the identified real object.

[0144] According to an example embodiment, at least one processor, individually and/or collectively, may be configured to cause the electronic device to display a virtual operation state corresponding to an operation state of the IoT device during operation within the image.

[0145] According to an example embodiment, at least one processor, individually and/or collectively, may be configured to cause the electronic device to: display at least one control item corresponding to the virtual IoT device and display the virtual operation state within the image, based on an input to the control item.

[0146] According to an example embodiment, at least one processor, individually and/or collectively, may be configured to cause the electronic device to: display, based on displaying a plurality of virtual IoT devices, a plurality of control items corresponding to the plurality of virtual IoT devices, respectively.

[0147] According to an example embodiment, at least one processor, individually and/or collectively, may be configured to cause the electronic device to: display an affordance item corresponding to the virtual IoT device and provide a purchase screen for purchasing an IoT device corresponding to the virtual IoT device, based on an input to the affordance item.

[0148] According to an example embodiment, the electronic device may further include a communication module comprising communication circuitry configured to support short-range wireless communication, and at least one processor, individually and/or collectively, may be configured to cause the electronic device to: recognize at least one adjacent real IoT device using

the communication module and identify the recognized real IoT device from the image.

[0149] According to an example embodiment, at least one processor, individually and/or collectively, may be configured to cause the electronic device to: display at least one control item corresponding to the real IoT device, and control the communication module to transmit a control signal instructing operation of the real IoT device, based on an input to the control item.

[0150] According to an example embodiment, at least one processor, individually and/or collectively, may be configured to cause the electronic device to: display, based on a real IoT device being added to be displayed within the image due to a change in a viewpoint of the camera, a control item corresponding to the added real IoT device and remove, based on a real IoT device displayed within the image being excluded therefrom due to a change in a viewpoint of the camera, a control item corresponding to the excluded real IoT device.

[0151] According to an example embodiment, at least one processor, individually and/or collectively, may be configured to cause the electronic device to: identify at least one real object included in the image through analysis of the image and determine at least one IoT device capable of being disposed in the at least one space, based on at least one of a size, distribution, or density of the identified real object.

[0152] A method for providing a virtual IoT environment of an electronic device according to various example embodiments of the disclosure may include: activating a camera to display an image acquired in real time through the camera, analyzing the image to determine at least one space within the image, identifying at least one Internet of Things (IoT) device capable of being disposed in the at least one space, and displaying at least one virtual IoT device corresponding to the at least one identified IoT device on the at least one determined space of the image.

[0153] According to an example embodiment, the method may further include displaying a virtual operation state corresponding to an operation state of the IoT device during operation within the image.

[0154] According to an example embodiment, the method may further include: displaying at least one control item corresponding to the virtual IoT device, and the displaying of the virtual operation state may include displaying the virtual operation state within the image, based on a user input to the control item.

[0155] According to an example embodiment, the method may further include: displaying an affordance item corresponding to the virtual IoT device and providing a purchase screen for purchasing an IoT device corresponding to the virtual IoT device, based on an input to the affordance item.

[0156] According to an example embodiment, the method may further include: recognizing at least one adjacent real IoT device and identifying the recognized real IoT device from the image.

[0157] The electronic device according to various embodiments may be one of various types of electronic devices. The electronic devices may include, for example, a portable communication device (e.g., a smartphone), a computer device, a portable multimedia device, a portable medical device, a camera, a wearable device, a home appliance, or the like. According to an embodiment of the disclosure, the electronic devices are not limited to those described above.

[0158] It should be appreciated that various embodiments of the present disclosure and the terms used therein are not intended to limit the technological features set forth herein to particular embodiments and include various changes, equivalents, or replacements for a corresponding embodiment. With regard to the description of the drawings, similar reference numerals may be used to refer to similar or related elements. It is to be understood that a singular form of a noun corresponding to an item may include one or more of the things, unless the relevant context clearly indicates otherwise. As used herein, each of such phrases as “A or B,” “at least one of A and B,” “at least one of A or B,” “A, B, or C,” “at least one of A, B, and C,” and “at least one of A, B, or C,” may include any one of, or all possible combinations of the items enumerated together in a corresponding one of the phrases. As used herein, such terms as “1st” and “2nd,” or “first” and

“second” may be used to simply distinguish a corresponding component from another, and does not limit the components in other aspect (e.g., importance or order). It is to be understood that if an element (e.g., a first element) is referred to, with or without the term “operatively” or “communicatively”, as “coupled with,” “coupled to,” “connected with,” or “connected to” another element (e.g., a second element), the element may be coupled with the other element directly (e.g., wiredly), wirelessly, or via a third element.

[0159] As used in connection with various embodiments of the disclosure, the term “module” may include a unit implemented in hardware, software, or firmware, or any combination thereof, and may interchangeably be used with other terms, for example, “logic,” “logic block,” “part,” or “circuitry”. A module may be a single integral component, or a minimum unit or part thereof, adapted to perform one or more functions. For example, according to an embodiment, the module may be implemented in a form of an application-specific integrated circuit (ASIC).

[0160] Various embodiments as set forth herein may be implemented as software (e.g., the program **140**) including one or more instructions that are stored in a storage medium (e.g., internal memory **136** or external memory **138**) that is readable by a machine (e.g., the electronic device **101**). For example, a processor (e.g., the processor **120**) of the machine (e.g., the electronic device **101**) may invoke at least one of the one or more instructions stored in the storage medium, and execute it, with or without using one or more other components under the control of the processor. This allows the machine to be operated to perform at least one function according to the at least one instruction invoked. The one or more instructions may include a code generated by a compiler or a code executable by an interpreter. The machine-readable storage medium may be provided in the form of a non-transitory storage medium. Wherein, the “non-transitory” storage medium is a tangible device, and may not include a signal (e.g., an electromagnetic wave), but this term does not differentiate between where data is semi-permanently stored in the storage medium and where the data is temporarily stored in the storage medium.

[0161] According to an embodiment, a method according to various embodiments of the disclosure may be included and provided in a computer program product. The computer program product may be traded as a product between a seller and a buyer. The computer program product may be distributed in the form of a machine-readable storage medium (e.g., compact disc read only memory (CD-ROM)), or be distributed (e.g., downloaded or uploaded) online via an application store (e.g., PlayStore™), or between two user devices (e.g., smart phones) directly. If distributed online, at least part of the computer program product may be temporarily generated or at least temporarily stored in the machine-readable storage medium, such as memory of the manufacturer's server, a server of the application store, or a relay server.

[0162] According to various embodiments, each component (e.g., a module or a program) of the above-described components may include a single entity or multiple entities, and some of the multiple entities may be separately disposed in different components. According to various embodiments, one or more of the above-described components may be omitted, or one or more other components may be added. Alternatively or additionally, a plurality of components (e.g., modules or programs) may be integrated into a single component. In such a case, according to various embodiments, the integrated component may still perform one or more functions of each of the plurality of components in the same or similar manner as they are performed by a corresponding one of the plurality of components before the integration. According to various embodiments, operations performed by the module, the program, or another component may be carried out sequentially, in parallel, repeatedly, or heuristically, or one or more of the operations may be executed in a different order or omitted, or one or more other operations may be added.

[0163] While the disclosure has been illustrated and described with reference to various example embodiments, it will be understood that the various example embodiments are intended to be illustrative, not limiting. It will be further understood by those skilled in the art that various changes in form and detail may be made without departing from the true spirit and full scope of the

disclosure, including the appended claims and their equivalents. It will also be understood that any of the embodiment(s) described herein may be used in conjunction with any other embodiment(s) described herein.

Claims

1. An electronic device comprising: a display; a camera; a memory configured to store instructions; and at least one processor, comprising processing circuitry, operatively connected to the display, the camera, and the memory, wherein at least one processor, individually and/or collectively, is configured to execute the instructions and to cause the electronic device to: activate the camera and display an image acquired in real time through the camera on the display; analyze the image to determine at least one space within the image; identify at least one Internet of Things (IoT) device capable of being disposed in the at least one space; and display at least one virtual IoT device corresponding to the at least one identified IoT device on the at least one determined space of the image.
2. The electronic device of claim 1, wherein at least one processor, individually and/or collectively, is configured to cause the electronic device to: identify at least one real object included in the image through analysis of the image; and identify an IoT device that is connectable with the identified real object.
3. The electronic device of claim 1, wherein at least one processor, individually and/or collectively, is configured to cause the electronic device to: display a virtual operation state corresponding to an operation state of the IoT device during operation within the image.
4. The electronic device of claim 1, wherein the at least one processor, individually and/or collectively, is configured to cause the electronic device to: display at least one control item corresponding to the virtual IoT device; and display the virtual operation state within the image, based on an input to the control item.
5. The electronic device of claim 4, wherein at least one processor, individually and/or collectively, is configured to cause the electronic device to: display, based on displaying a plurality of virtual IoT devices, a plurality of control items corresponding to the plurality of virtual IoT devices, respectively.
6. The electronic device of claim 4, wherein at least one processor, individually and/or collectively, is configured to cause the electronic device to: display the image acquired through the camera on a first area of the display; and display the at least one control item on a second area of the display.
7. The electronic device of claim 1, wherein at least one processor, individually and/or collectively, is configured to cause the electronic device to: display an affordance item corresponding to the virtual IoT device; and provide a purchase screen for purchasing an IoT device corresponding to the virtual IoT device, based on an input to the affordance item.
8. The electronic device of claim 1, further comprising a communication module comprising communication circuitry configured to support short-range wireless communication, wherein at least one processor, individually and/or collectively, is configured to cause the electronic device to: recognize at least one adjacent real IoT device using the communication module; and identify the recognized real IoT device from the image.
9. The electronic device of claim 1, wherein at least one processor, individually and/or collectively, is configured to cause the electronic device to: display at least one control item corresponding to the real IoT device; and transmit a control signal instructing operation of the real IoT device, based on an input to the control item.
10. The electronic device of claim 8, wherein at least one processor, individually and/or collectively, is configured to cause the electronic device to: display, based on a real IoT device being added to be displayed within the image due to a change in a viewpoint of the camera, a control item corresponding to the added real IoT device; and remove, based on a real IoT device

displayed within the image being excluded therefrom due to a change in a viewpoint of the camera, a control item corresponding to the excluded real IoT device.

11. The electronic device of claim 10, wherein at least one processor, individually and/or collectively, is configured to cause the electronic device to: identify at least one real object included in the image through analysis of the image; and determine at least one IoT device capable of being disposed in the at least one space, based on at least one of a size, distribution, or density of the identified real object.

12. A method for providing a virtual IoT environment of an electronic device, the method comprising: activating a camera to display an image acquired in real time through the camera; analyzing the image to determine at least one space within the image; identifying at least one Internet of Things (IoT) device capable of being disposed in the at least one space; and displaying at least one virtual IoT device corresponding to the at least one identified IoT device on the at least one determined space of the image.

13. The method of claim 12, further comprising displaying a virtual operation state corresponding to an operation state of the IoT device during operation within the image.

14. The method of claim 13, further comprising displaying at least one control item corresponding to the virtual IoT device, wherein the displaying of the virtual operation state comprises displaying the virtual operation state within the image, based on an input to the control item.

15. The method of claim 14, wherein displaying of the at least one control item comprises: displaying, based on displaying a plurality of virtual IoT devices, a plurality of control items corresponding to the plurality of virtual IoT devices, respectively.

16. The method of claim 12, further comprising: displaying an affordance item corresponding to the virtual IoT device; and providing a purchase screen for purchasing an IoT device corresponding to the virtual IoT device, based on an input to the affordance item.

17. The method of claim 12, further comprising: recognizing at least one adjacent real IoT device; and identifying the recognized real IoT device from the image.

18. The method of claim 12, further comprising: displaying at least one control item corresponding to the real IoT device; and transmitting a control signal instructing operation of the real IoT device, based on an input to the control item.

19. The method of claim 18, further comprising: displaying, based on a real IoT device being added to be displayed within the image due to a change in a viewpoint of the camera, a control item corresponding to the added real IoT device; and removing, based on a real IoT device displayed within the image being excluded therefrom due to a change in a viewpoint of the camera, a control item corresponding to the excluded real IoT device.

20. A non-transitory computer-readable recording medium storing instructions, when executed by at least one processor, to perform a method comprising: activating a camera to display an image acquired in real time through the camera; analyzing the image to determine at least one space within the image; identifying at least one Internet of Things (IoT) device capable of being disposed in the at least one space; and displaying at least one virtual IoT device corresponding to the at least one identified IoT device on the at least one determined space of the image.
